Current makes a magnetic field

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “magnetic field” to explain your model or result. Sentence starters are optional.

current: rate of charge flow

magnetic field: a field that influences magnets and moving charges

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A total current of 12 A splits between two identical parallel branches.
Find each branch current.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ T·m/A.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and explain what the dot represents.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Current out of page (dot). Add field arrows.



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Conventional current points out of the page.
Use the right-hand rule to draw the surrounding magnetic field direction.

Sentence starter: The … in my model represents … . Its label shows … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A long straight wire carries current out of the page. At a fixed distance, the current doubles.
Use B ∝ I to compare field strengths. Add field arrows and indicate whether their direction changes.

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

05 Explain your reasoning

What happens to field direction if the current reverses while its magnitude stays the same?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

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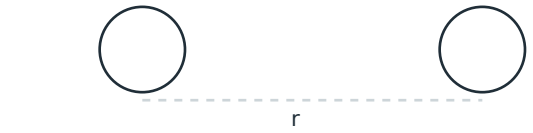
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Two objects have charges -4 nC and -6 nC.
Predict attraction or repulsion and draw both force directions.
Label charges and both interaction forces.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

07 Exit check

At the same current, distance from a long wire doubles. Using B proportional to 1/r, compare field strength.

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Reverse the current. Redraw the field arrows and explain what stays the same in an ideal symmetric setup.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Cross = current into page. Add magnetic-field arrows.



Marzano check - circle: 1 2 3